

Praveena Kurukuladithya

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OBJECTIVE

Computer Systems Engineering undergraduate with hands-on experience in **embedded systems, Industrial IoT, robotics, PLC automation, PCB design, and Linux-based development**. Experienced in building hardware-software systems using **ESP32, Arduino, Raspberry Pi, sensors, motor drivers, MQTT, Flask/FastAPI dashboards, and control algorithms**. Seeking an **Embedded Systems, IoT, Robotics, or Industrial Automation internship** to apply practical engineering skills in real-world product development, testing, troubleshooting, and deployment.

TECHNICAL SKILLS

Embedded & IoT: ESP32, ESP32-S3, Arduino, Raspberry Pi, FreeRTOS, GPIO, UART, I2C, SPI, RS485, PWM, ADC, Timers, Interrupts, MQTT, WebSocket, Sensor Interfacing

Robotics & Control: PID Control, DC Motor Control, Encoder Feedback, Motor Drivers (TB6612FNG, L298N), Line Following, Obstacle Detection, Real-Time Control Logic, AVR Assembly

PCB & Electronics: EasyEDA, JLCPCB, PCB Design, Custom PCB Carrier Boards, PCB Etching, Schematic Design, Soldering, Circuit Debugging, Multimeter Testing, Power Distribution

Languages: C, C++, Python, JavaScript, HTML/CSS, Bash, AVR Assembly, SQL

Web & Cloud: React.js, Node.js, Express.js, FastAPI, Flask, Supabase, PostgreSQL, Supabase Realtime, Firebase, REST APIs, Chart.js, Vercel, MERN Stack, Git, GitHub

Industrial Automation: Siemens S7-200, Siemens S7-1200, PLC Programming, PLC Wiring, Ladder Logic, Automation Troubleshooting

Linux & DevOps: Raspberry Pi OS, Linux Terminal, SSH, Docker, Docker Compose, InfluxDB, Mosquitto MQTT Broker, Web App Deployment

PROJECTS

EdgePulse IIoT — Industrial Motor Health Monitoring & PLC Control

Apr. 2026 – Present

- **Tech:** ESP32-S3, C++, FreeRTOS, Arduino UNO, RS485, MQTT, Raspberry Pi, InfluxDB, FastAPI, React, Docker Compose, Custom PCB (EasyEDA/JLCPCB)
- Architected a **3-layer industrial IIoT prototype** Arduino UNO PLC layer (TB6612FNG + INA3221), ESP32-S3 intelligent embedded layer (C++/FreeRTOS), and Raspberry Pi edge server for real-time motor/fan condition monitoring and remote PLC-style control.
- Implemented **FreeRTOS multi-task firmware** for concurrent ADXL345 vibration, INMP441 acoustic, and motor electrical sensing; published telemetry via **MQTT over RS485** to Mosquitto broker.
- Built **Raspberry Pi edge stack** (FastAPI + InfluxDB + React + Docker Compose) for live dashboards and time-series storage; designed **custom PCB carrier boards** for both embedded layers.

LittleGurd — Cloud-Connected Smart Bag IoT Monitoring System

Ongoing

- **Tech:** React.js, Vite, Supabase, PostgreSQL, Supabase Realtime, Python, Tkinter, Vercel | Planned: ESP32/STM32, GPS, 4G
- Developed a **cloud-connected IoT monitoring system** tracking GPS, weight, zip/wearing status, battery, and safety alerts via a React dashboard with **Supabase Realtime** zero-polling live telemetry.
- Built a **Python Tkinter embedded simulator** emulating ESP32/STM32 IoT device behavior to stream structured sensor data to the cloud for full end-to-end testing before physical hardware.
- Implemented **multi-bag profile management** with Bag ID generation, online/offline detection, and auto-disconnect handling; designed hardware roadmap for GPS, 4G, and sensor integration.

PID Motor Control System & IIoT SCADA — Raspberry Pi, Python, Flask

- **Tech:** Raspberry Pi, Python, Flask, Chart.js, TB6612FNG, 700-PPR Optical Encoder, PWM, DSP, pandas, scipy
- Built a **closed-loop DC motor speed control system** with empirical transfer function derivation via open-loop step response and **PID control with anti-windup** and real-time PWM output on Raspberry Pi.
- Developed a **Flask-based SCADA dashboard** (Chart.js) for live RPM monitoring and dynamic PID tuning; applied **DSP moving-average filtering** to eliminate encoder quantization noise.

Autonomous Line Following, Obstacle Detection & Parking Robot — AVR Assembly

- **Tech:** Arduino Uno, ATmega328P, AVR Assembly, IR Sensors, Ultrasonic Sensor, TB6612FNG, Timer0 PWM
- Developed a **fully autonomous robot in AVR Assembly** with priority-based control logic obstacle detection overrides line-following, triggering safe stop and automatic path resumption.
- Designed a **state-machine** for line following, correction, parking-zone detection, and auto-parking sequence; controlled speed via **Timer0 PWM** and direction via ATmega328P digital I/O.

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT)

BSc (Hons) in Computer Systems Engineering

Nov. 2023 – Present

Malabe, Sri Lanka

- **Relevant Areas:** Embedded Systems, Control Systems, Digital Systems, DSP, Electronics, IoT, Robotics, Programming
- **Current GPA:** 2.80

Kalutara Vidyalaya

GCE Advanced Level — Physical Science Stream | Z-Score: 0.6896

2008 – 2022

Kalutara, Sri Lanka

CERTIFICATIONS & QUALIFICATIONS

Designing, Maintaining & Troubleshooting Automated Engineering Systems — PLC

Nov. 2025 – Feb. 2026

Robotics & Control Systems Lab, Dept. of Mechanical Eng., Faculty of Engineering

University of Moratuwa, Sri Lanka

- Completed hands-on training in **PLC programming, PLC wiring, industrial automation troubleshooting, and automated system maintenance.**
- Practiced **Siemens S7-200 and S7-1200 PLC programming** with real industrial automation wiring and control scenarios.

LEADERSHIP & ACTIVITIES

Batch Representative — Computer Systems Engineering Batch 02

Jan. 2024 – Present

Department of Computer Systems Engineering, SLIIT

Malabe, Sri Lanka

- Represent the batch and coordinate academic communication, student concerns, department updates, and event-related announcements.

Executive Committee Member

Aug. 2025 – Present

Computer Systems Engineering Student Committee, SLIIT

Malabe, Sri Lanka

- Support planning and coordination of department-level student activities, technical events, and community engagement initiatives.

Event Coordinator / Committee Member

2024 – Present

IdeaSpark 2 & 3, AGMs, CodeFest 2025, Innova IoT

Sri Lanka

- Contributed to event planning, communication, participant coordination, and technical/community engagement activities.

SOFT SKILLS

Problem Solving | Practical Debugging | Teamwork | Leadership | Communication | Time Management | Self-Learning | Adaptability | Technical Documentation | Project Coordination

REFERENCES

Prof. Anuradha Jayakody

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